SUMMARY REPORT

FOR

Mon Logistic

ON

THE APPLICATION OF 3M SOLAR CONTROL FILM FOR CLEAR GLASS

Prepared by

Pichayapong Boonlue

Sr. Application Engineer-CSD Film

Customer Technical Center, 3M Thailand Co., Ltd

Customer-Applicator Information

Customer Name: Mon Logistic

Address : Bangkok

Name : K. Sitthisak
Phone Number : 063-2307390
E-mail :

Dealer Name : Good film
Address : Bangkok
Contact Name : K. Kanin Siriboonthanasook
Phone Number : 063-6242391

Site Information

Glass Type : Clear glass
Glass Size : Est. 0.8 metre × 1.2 metre (Width × Height)
Glass Thickness: Clear Glass (6 mm. Thickness)
Proposed Solar Control Films Types: 3M BC20, NV25

Specification of 3M BC20 and NV25

Product Performance & Technical Data

BC20	Single Pane		Single Pane Auto75	
Film	1/4" Clear	BC20	1/4" Auto75	BC20
Solar Heat Gain Coefficient	0.82	0.32	0.59	0.32
Visible Light Transmitted	89%	17%	75%	14%
Visible Light Reflected Interior	9%	25%	7%	25%
Visible Light Reflected Exterior	8%	20%	7%	15%
U Value	1.03	0.97	1.02	0.97
UV Block	38%	99%	NA	99.9%
Total Solar Energy Rejected	19%	68%	41%	68%
Glare Reduction	NA	81%	NA	81%
Heat Loss Reduction	NA	6%	NA	5%
Solar Heat Reduction	NA	60%	NA	46%

Product Performance & Technical Data

NV25	Single Pane		Tinted		Double Pane		Double tinted	
Film	1/4" Clear	NV25	1/4" tint	NV25	Dual 1/4" Clear	NV25	Dual 1/4" tint	NV25
Solar Heat Gain Coefficient	0.82	0.39	0.63	0.36	0.70	0.49	0.51	0.37
Visible Light Transmitted	89%	24%	53%	14%	79%	22%	47%	13%
Visible Light Reflected Interior	9%	7%	6%	7%	15%	8%	13%	8%
Visible Light Reflected Exterior	8%	19%	6%	10%	15%	24%	8%	12%
U Value	1.03	1.00	1.03	1.00	0.47	0.47	0.47	0.47
UV Block	38%	99%	NA	99%	NA	99%	NA	99%
Total Solar Energy Rejected	19%	61%	37%	64%	30%	51%	49%	63%
Glare Reduction	NA	73%	NA	74%	NA	72%	NA	73%
Heat Loss Reduction	NA	2%	NA	2%	NA	1%	NA	1%
Solar Heat Reduction	NA	52%	NA	43%	NA	29%	NA	27%

Objective of the Test

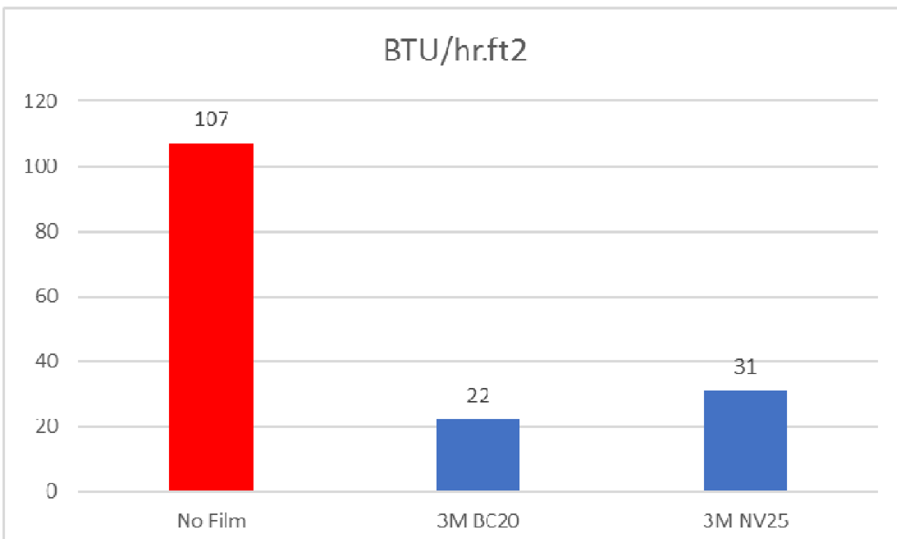
To clarify the performance of 3M Solar Control films 3M BC20, NV25 and No film by considering the glass with Temperature different by using temperature logger and BTU Meter.

Scope of the Test

1. Test Duration: This test was conducted from Mar27, 2019 to Apr2,2019
2. Test Position: This test was conducted inside building Panes of glass windows were selected to conduct the experiment in order to compare the difference of heat transfer through glass windows between glass window with 3M BC20, NV25 and No film

Test Result

1) Solar Energy through the window.



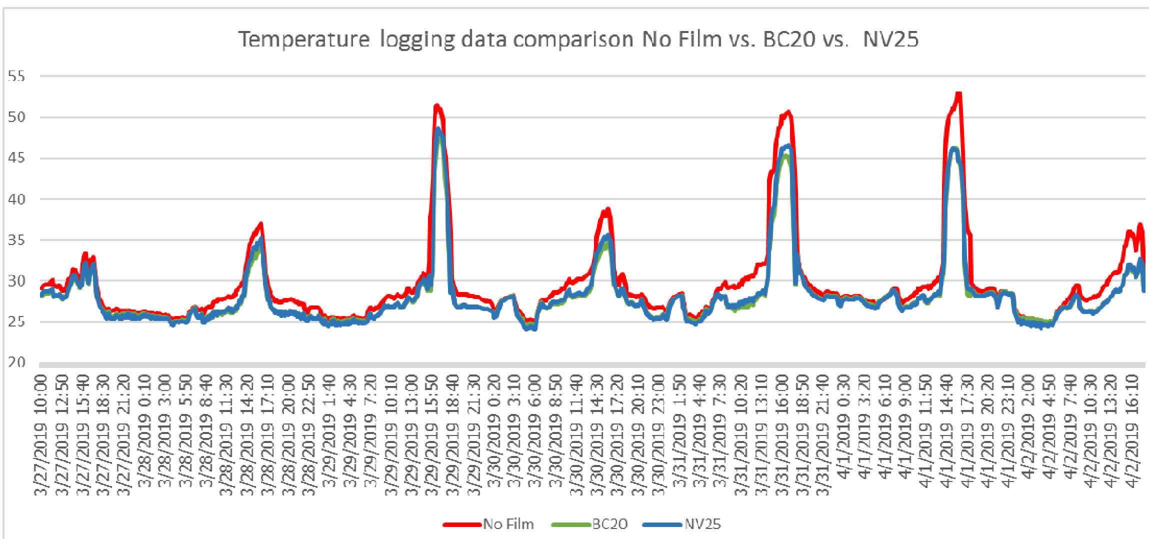
Solar Energy measured by BTU meter

No Film

3M BC20 3M NV25

Solar Energy Measured by BTU meter

2) Temperature logging data



Record temperature by Temp logger

Date & Time	No Film	BC20	Diff	NV25	Diff
3/27/2019 16:10	33.3	31.9	1.4	32.3	1
3/28/2019 16:20	37	34.6	2.4	35.3	1.7
3/29/2019 16:40	51.4	46.9	4.5	48.3	3.1
3/30/2019 16:10	38.8	34.5	4.3	35.6	3.2
3/31/2019 17:00	50.7	45	5.7	46.5	4.2
4/1/2019 16:20	52.9	45.5	7.4	45.4	7.5
4/2/2019 17:30	36.9	32.5	4.4	32.7	4.2

Maximum temperature of each day

Conclusion

1)The results of the solar energy after apply film measured by Solar Transmission meter.

The Solar energy reduced from 107 to 22 BTU/Hr.ft^2 for 3M BC20

The Solar energy reduced from 107 to 31 BTU/Hr.ft^2 for 3M NV25

2)The maximum different temperature of each day is 7.5 Degree C on Apr1,2019

Remark The test result depends on many factors including factors from outside condition and inside condition such as the environmental condition in each day, raining season, human factors at the experimental location, performance of air-conditioning system, Size of film sample that apply on glass etc. Some tolerance between data from field testing and the data from laboratory testing may be occurred during observation. However, the results from this experiment are conducted between the samples of window films only which may get the lower performance than the results from the application to whole building after apply film.

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